

SKILLS

- **Risk/Data Pipelines:** Python, SQL, Pandas, ETL/ELT, trade ingestion, normalization, reconciliation, P&L analytics
- **Financial Systems:** risk controls, position/inventory monitoring, exposure analytics, trade booking, regulatory feeds
- **Market/Trade Data:** Kafka, Airflow, streaming pipelines, batch processing, data validation, lineage, replayability
- **Databases:** PostgreSQL, MySQL, Redis, Elasticsearch, Snowflake, schema design, query tuning, data modelling
- **Integrations:** REST APIs, HTTP, file feeds, vendor data, audit logs, secure routing, FIX protocol concepts
- **Cloud/Infra:** AWS, S3, Lambda, EKS, Docker, Kubernetes, Terraform, CI/CD, Linux, GitHub Actions
- **Reliability:** OpenTelemetry, ELK, CloudWatch, monitoring, alerting, SLOs, runbooks, level-2 support
- **Leadership:** requirements capture, architecture design, bottleneck analysis, code reviews, mentoring, demos

EXPERIENCE

- Royal Bank of Canada (RBC)

Staff Software Developer & Enterprise Architect – Data / Risk Platform

Toronto, Canada

Nov 2022 – Present

 - **High-Scale Risk-Style Data Pipelines:** Architected telemetry platform using OpenTelemetry, Vector, Fluent Bit, Kafka, and Logstash ingesting 150 TB/day across 10,000+ systems; standardised schemas, transformations, routing, validation, alerting, replayable ingestion, and downstream analytics.
 - **Real-Time Monitoring & Controls:** Designed production controls for critical platform data flows including readiness checks, traceability, SLOs, anomaly detection, incident triage, escalation paths, runbooks, and level-2 support workflows for regulated systems.
 - **Trade-Like Data Normalization:** Built reusable ingestion and integration patterns to normalize heterogeneous enterprise events into governed schemas, reduce point-to-point vendor feeds, and make downstream analytics queryable and auditable across teams.
 - **Stakeholder Requirements & Architecture:** Partner with architecture, product, infrastructure, security, and risk teams to capture requirements, define architecture and technologies, identify bottlenecks, demo functionality, and drive improvements after production launch.
 - **Technical Leadership:** Led 5 engineers through design discussions, code reviews, testing expectations, ADRs, documentation, mentoring, and reusable platform standards adopted by 20+ engineering teams.
- HealthCard (Acquired Startup)

Blockchain Architect & First Full Stack Engineer

Remote

Aug 2021 – Jul 2022

 - **Financial Workflow Automation:** Joined as first engineer and built React/TypeScript frontend, Node.js/Express APIs, AWS EKS deployment, audit logging, verification workflows, and trust-score data models for real-time premium validation and claims processing.
 - **Reconciliation & Validation Pipelines:** Designed insurance validation logic and document classification workflows with monitoring, feedback loops, and HIPAA-aligned audit trails; reduced manual review by 60% while improving correctness of credential verification.
- CryptoVantage

Blockchain Developer

Netherlands (Remote)

Jul 2020 – Dec 2020

 - **Real-Time Risk Scoring:** Built on-chain analytics in Go/JavaScript with Ethereum event streaming, wallet clustering, and risk scoring for AML-style transaction monitoring and suspicious-pattern detection.
- Dalhousie University

Research & Teaching Assistant

Halifax, Canada

Jan 2021 – Aug 2022

 - **Algorithms, Systems & Cloud:** Researched call-stack decision algorithms and Python tracing tools for compiler/runtime analysis; TA for Algorithms, Software Development, and graduate Advanced Cloud Computing.

PROJECTS

- **Trade Reconciliation Pipeline:** Python prototype for normalizing broker/trade-like feeds, validating booking fields, reconciling positions and P&L deltas, and producing exception reports with replayable audit logs.
- **Automatic Trading Algorithm:** Sub-second latency personal project integrating live brokerage APIs and real-time market data streams; momentum/mean-reversion signals with risk controls, position sizing, and performance monitoring.
- **Portfolio Risk Optimization Framework:** Python research system for market data ingestion, portfolio allocation, Sharpe analysis, Monte Carlo simulation, risk-adjusted signals, and repeatable quantitative experiments.

EDUCATION

- Dalhousie University

MASc Applied Computer Science, GPA: 3.9/4.0 | TA: Algorithms, Advanced Cloud Computing

Halifax, Canada

Jan 2021 – Aug 2022
- University of Toronto

Enterprise Architecture Certification (3-semester program)

Toronto, Canada

Completed 2024
- Gujarat Technological University

B.E. Computer Engineering, GPA: 3.92/4.0

Ahmedabad, India

Aug 2016 – Aug 2020